

Shashank Raj

Lead Software Engineer | GenAI & ML

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DevTrack (Open Source): devtrack.cloud — Local-first developer automation platform; converts a problem statement into a fully planned sprint, syncs commits to Azure DevOps/GitLab/GitHub in real time, full control via Telegram.

runtime-narrative (PyPI): pypi.org/project/runtime-narrative — Open-source Python library that models application execution as human-readable stories, replacing raw tracebacks with structured stage-by-stage failure reports; supports FastAPI middleware, JSON streaming, and LLM-powered failure analysis.

SUMMARY

Results-oriented Data Scientist and AI Engineer with 14+ years of experience specializing in building intelligent systems, scalable ML workflows, and production-grade AI pipelines. Expert in deploying agentic frameworks, generative AI, and RAG architectures to transform complex data into actionable business insights, driving efficiency gains of up to 85% through automation and predictive modeling.

TECHNICAL SKILLS

Data Science & ML TensorFlow, scikit-learn, XGBoost, LightGBM, MLflow, Neural Networks, NLP, MLOps, KServe, Kubeflow, Predictive Analytics

AI & LLM Technologies GPT-4, Claude, Azure OpenAI, Ollama, LangChain, LangGraph, LangSmith, Mixtral, ChromaDB, Multi-Agent Orchestration, RAG Pipelines, Prompt Engineering, Agentic AI (MCP, A2A)

Cloud Technologies AWS (Athena, S3, Redshift, Lambda), Databricks, Sagemaker, GitHub Actions, MLOps Pipelines

Programming & Frameworks Python, Golang, SQL, SQLAlchemy, Pandas, PyTorch, FastAPI, Docker, Kubernetes, Apache Airflow, Apache Spark, Kafka, Git, Linux

WORK EXPERIENCE

Lead Software Engineer

EPAM Systems PVT. LTD. | Pune, Maharashtra

August 2025 – Present

Client: S&P Global Ratings

- Developed and maintained production-grade Python code supporting MLOps/LLMOps workflows across model training, deployment, and monitoring.
- Designed and implemented data solutions using Elasticsearch, Apache Airflow, Apache Spark, Kafka, Databricks, Sagemaker, and MLflow for efficient model lifecycle management.
- Built AI-driven applications using LangChain, LangGraph, and LangSmith; designed and deployed Agentic AI architectures incorporating MCP and Google A2A protocols.
- Developed and integrated RAG pipelines with prompt engineering to solve complex business challenges using Generative AI.
- Managed containerized Kubernetes deployments on cloud platforms; established CI/CD pipelines for streamlined development and release.
- Architected distributed microservices systems for large-scale batch and stream processing analytics pipelines.

Key Projects:

- VeriShield:** AI-powered KYC and BGV verification platform minimizing manual intervention and accelerating onboarding. Role: Solution Architect, Product Manager, End-to-End Developer. Selected in top 10 at EPAM AI Hackathon, cleared for customer POC.
- RAVEN:** Unified security and observability platform for GenAI solutions, integrating LLM guardrails, vulnerability scanning, and compliance monitoring. Role: Solution Architect, Product Manager.
- AI Cosmos:** Internal AI knowledge-sharing platform for the NRG + EU Delivery Unit, enabling colleagues to publish articles on AI concepts, implementations, and use cases. Role: Solution Architect, Product Manager, Backend Developer.

Senior Manager, Data Science

LatentBridge | Pune, Maharashtra

April 2025 – July 2025

- Architected a production-ready natural language to SQL engine for client-facing applications, reducing research time from 4+ hours to minutes and boosting query efficiency by 85% through intelligent metadata preprocessing and adaptive caching.
- Engineered an agentic AI research application with multi-LLM orchestration (GPT-4, Azure OpenAI, Ollama) and LangGraph, cutting manual investigation time by 80%.
- Implemented NLTK and SpaCy-based intent recognition engines, improving query accuracy by 40% and optimizing human-AI task distribution.
- Designed scalable agentic RAG frameworks with Mixtral and Azure vector embeddings, enhancing information extraction efficiency by 45% across business functions.

Lead Data Engineer

Daaslabs.ai | Gurgaon, Haryana

February 2024 – February 2025

- Designed a scalable ETL system using Python and PySpark, processing 500+ GB of daily data and improving throughput by 50% to enable real-time AI model training.
- Built a text-to-SQL prototype for complex enterprise schemas, with metadata caching that increased query accuracy by 35%.
- Implemented an MLflow-based model deployment framework with versioning, reducing deployment downtime by 40%.
- Optimized data transformation workflows with caching and parallel processing, improving computational efficiency by 45%.

Senior Manager, Data & Analytics

Exalogic Consulting | Dubai, UAE

January 2023 – January 2024

- Developed advanced time series forecasting and enterprise planning algorithms, reducing insight generation time from one week to minutes (75% efficiency gain).
- Engineered a cloud-agnostic migration strategy to Kubernetes and Docker microservices, enhancing scalability by 30%.
- Built integration engines for automated data workflows, reducing ML deployment time by 35%.
- Implemented GDPR/CCPA-compliant data governance frameworks, reducing compliance risks by 25%.

Senior Analyst, Run and Support

Syngenta Service PVT LTD. | Pune, Maharashtra

February 2021 – January 2023

- Orchestrated MLOps migration to KServe and Kubeflow, saving millions in infrastructure costs and improving operational efficiency by 30% through advanced hyperparameter tuning.
- Developed data models for agricultural tech initiatives, improving decision-making reliability and scaling ML systems by 40%.
- Established ML monitoring frameworks with automated scripts, reducing production errors by 30%.
- Led cross-functional collaboration on data-driven agricultural solutions, saving 10+ weekly hours through 50% automation.

PREVIOUS EXPERIENCE

Senior Packaging Consultant / Application Developer | IBM (Global Business Development) | May 2015 – January 2021

SAP BI Architect | MResult | November 2013 – June 2015

Software Engineer | KPIT | January 2011 – October 2013

EDUCATION

M.Sc. in Business Analytics

2021

Birla Institute of Technology and Science, Pilani (BITS Pilani)

Bachelor of Computer Applications

2011

Indira Gandhi National Open University (IGNOU) [UGC, AICTE, COL, DEC, AIU]